Mock diagnostic examination - Probability and Statistics paper

Candidate paper

Time allowed: 90 minutes
Marks for this paper: 45
Marks across both mock papers: 100
Questions: All questions and all parts are compulsory.

Instructions

- Complete this paper independently of the Maths paper.
- Complete the paper without notes or solutions. You may use the supplied Probability and Statistics formula sheet.
- A basic non-programmable scientific calculator is permitted.
- No computational or statistical software is needed or permitted. Approved assistive technology, accessibility accommodations and adjusted timing remain permitted where authorised; assistive tools must not perform the assessed probability or statistics.
- Show the reasoning and intermediate work needed to support each answer. An unsupported answer may not receive full marks.
- Give exact answers unless the question asks for an approximation. Round requested decimal answers to four decimal places unless stated otherwise.
- State relevant supports, parameter restrictions and conditions when they affect a method or conclusion.

Section S0 — short diagnostic questions (10 marks; 15 minutes)

Question 1 — `MOCK-S0-Q01-PA` (2 marks; 3 minutes) A two-person committee is selected from five economists and three statisticians. How many different committees contain exactly one economist and one statistician?

Question 2 — `MOCK-S0-Q02-PA` (2 marks; 3 minutes) Let $X \sim \text{Exponential}(2)$, where 2 is the rate. State the support of X and calculate $\mathbb{P}(X > 1)$ exactly.

Question 3 — `MOCK-S0-Q03-PA` (2 marks; 3 minutes) Let $X_1, \dots, X_{16}$ be iid with

$$\mathbb{E}[X_i] = \mu, \qquad \text{Var}(X_i) = 4.$$

Calculate $\mathbb{E}[\bar{X}]$ and $\text{Var}(\bar{X})$.

Question 4 — `MOCK-S0-Q04-PA` (2 marks; 3 minutes) Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda), \qquad \lambda > 0,$$

and let $\hat{\lambda} := \bar{X}$. For the observed sample $(1, 0, 2, 1)$, identify the parameter, the estimator and the numerical estimate.

Question 5 — `MOCK-S0-Q05-PA` (2 marks; 3 minutes) A two-sided hypothesis test returns a p-value of 0.08. State the decision at the 5% significance level, and explain why this result does not prove that the null hypothesis is true.

Section S1 — probability, random variables and joint distributions (15 marks; 30 minutes)

Question 6 — total probability and Bayes' rule (5 marks; 10 minutes)

Part A — `MOCK-S1-Q06-PA` A student follows revision route A with probability $3/5$, and follows the alternative route A^c otherwise. Let D be the event that the student completes a diagnostic task. Suppose

$$\mathbb{P}(D \mid A) = \frac{3}{4}, \qquad \mathbb{P}(D \mid A^c) = \frac{1}{2}.$$

Calculate $\mathbb{P}(D)$ and $\mathbb{P}(A \mid D)$. Interpret the latter probability among students who complete the task.

Question 7 — a joint probability mass function (10 marks; 20 minutes) The joint probability mass function of X and Y is given in the following table. Entries not displayed have probability zero.

$p_{X,Y}(x,y)$	$y=0$	$y=1$	$y=2$
$x=0$	$2/12$	$1/12$	$1/12$
$x=1$	$1/12$	$3/12$	$4/12$

Part A — `MOCK-S1-Q07-PA` (5 marks; 10 minutes) Verify that the table defines a valid joint PMF. Derive the marginal PMFs of X and Y, then calculate $\mathbb{E}[X]$ and $\mathbb{E}[Y]$.

Part B — `MOCK-S1-Q07-PB` (5 marks; 10 minutes) Derive the conditional PMF $p_{Y|X}(y \mid 1)$, including its support, and calculate $\mathbb{E}[Y \mid X = 1]$. Determine whether X and Y are independent, giving one numerical justification.

Section S2 — sampling distributions, the LLN and the CLT (8 marks; 15 minutes)

Question 8 — exact and approximate sample-mean calculations (8 marks; 15 minutes)

Part A — `MOCK-S2-Q08-PA` (4 marks; 7 minutes) Let

$$X_1, \dots, X_9 \overset{\text{iid}}{\sim} \mathcal{N}(12, 9).$$

State the exact distribution of $\bar{X}$, calculate $\mathbb{P}(\bar{X} < 13)$, and explain why the distributional calculation is exact rather than an approximation.

Part B — `MOCK-S2-Q08-PB` (4 marks; 8 minutes) Let $W_1, \dots, W_{100}$ be iid random variables with

$$\mathbb{E}[W_i] = 50, \qquad \operatorname{Var}(W_i) = 100.$$

Use the central limit theorem to approximate $\mathbb{P}(48 < \bar{W} < 52)$. State the conditions used by the approximation, then explain what the law of large numbers says about $\bar{W}$ as the sample size increases.

Section S3 — estimation and inference (12 marks; 30 minutes)

Question 9 — estimator properties and method of moments (5 marks; 12 minutes)

Part A — `MOCK-S3-Q09-PA` Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda), \qquad \lambda > 0.$$

Derive the one-moment method-of-moments estimator of λ. Compare the two estimators

$$T_1 = \bar{X} \qquad \text{and} \qquad T_2 = X_1$$

by calculating their biases and variances. Use these results to state which estimator is consistent for λ, with justification.

Question 10 — a confidence interval and a two-sided test (7 marks; 18 minutes)

Part A — `MOCK-S3-Q10-PA` A sample of size $n = 16$ from a population with unknown mean μ and unknown variance gives

$$\bar{x} = 52.2, \qquad s = 4.$$

Use an exact one-sample Student's t procedure. State the sampling and population-shape conditions needed for this procedure.

1. Test $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$ at the 5% significance level. Calculate the test statistic, use the supplied two-sided p-value and state the decision.
2. Construct the corresponding 95% confidence interval for μ.
3. Explain why the interval and test decisions agree. Give a valid repeated-sampling interpretation of the confidence procedure.

Formula sheet

This sheet accompanies the 90-minute, 45-mark Probability and Statistics mock paper.

General conventions

- Random variables use upper-case letters and their realisations use lower-case letters.
- Use $p_X(x)$ for a PMF, $f_X(x)$ for a PDF and $F_X(x) = \mathbb{P}(X \leq x)$ for a CDF.
- The normal distribution is parameterised as $\mathcal{N}(\mu, \sigma^2)$, where the second parameter is the variance.
- The exponential parameter λ is a rate, and a geometric random variable counts trials up to and including the first success.
- For a random sample,

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Standard distributions

In the table below, a PMF or PDF is zero outside the stated support.

Distribution	PMF or PDF and support	Mean	Variance
$X \sim \text{Bernoulli}(p)$, $0 \leq p \leq 1$	$p_X(1) = p$, $p_X(0) = 1-p$, $x \in \{0,1\}$	p	$p(1-p)$
$X \sim \text{Binomial}(n,p)$, $n \in \{1,2,...\}$, $0 \leq p \leq 1$	$p_X(x) = \binom{n}{x}p^x(1-p)^{n-x}$, $x \in \{0,\dots,n\}$	np	$np(1-p)$
$X \sim \text{Geometric}(p)$, $0 < p \leq 1$	$p_X(x) = p(1-p)^{x-1}$, $x \in \{1,2,...\}$	$1/p$	$(1-p)/p^2$
$X \sim \text{Poisson}(\lambda)$, $\lambda > 0$	$p_X(x) = e^{-\lambda}\lambda^x/x!$, $x \in \{0,1,2,...\}$	λ	λ

Distribution	PMF or PDF and support	Mean	Variance
$X \sim \text{Uniform}(a, b)$, $a < b$	$f_X(x) = 1/(b-a)$, $a \le x \le b$	$(a+b)/2$	$(b-a)^2/12$
$X \sim \text{Exponential}(\lambda)$, $\lambda > 0$	$f_X(x) = \lambda e^{-\lambda x}$, $x \ge 0$	$1/\lambda$	$1/\lambda^2$
$X \sim \mathcal{N}(\mu, \sigma^2)$, $\sigma^2 > 0$	$f_X(x) = (\sigma\sqrt{2\pi})^{-1} \exp[-(x-\mu)^2/(2\sigma^2)]$, $x \in \mathbb{R}$	μ	σ^2

One-sample Student's t procedure

For an iid sample from a normal population with unknown variance, the two-sided $(1-\alpha)100\%$ confidence interval for μ is

$$\bar{X} \pm t_{1-\alpha/2,n-1} \frac{S}{\sqrt{n}}.$$

For testing $H_0 : \mu = \mu_0$, the test statistic is

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}},$$

which has a t_{n-1} distribution under H_0 when the stated conditions hold.

Supplied numerical values

Let Φ denote the standard normal CDF. The supplied decimal values are rounded to four decimal places. You may use

$$\Phi(1) \approx 0.8413, \qquad \Phi(2) \approx 0.9772, \qquad \Phi(-z) = 1 - \Phi(z).$$

For 15 degrees of freedom, you may use

$$t_{0.975,15} \approx 2.131.$$

For a two-sided t_{15} test, the p-value corresponding to $|t| = 2.20$ is approximately 0.0439.